Moniha R

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.81
Velammal Vidhyashram	Class XII (CBSE)	2024	94.00%
Jain Vidyaashram	Class X (CBSE)	2022	93.60%

SKILLS

Languages: Python, C, C++, SQL

Frameworks/Libraries: NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn, LangChain, LangGraph, Pydantic AI, Streamlit

Technologies: Git/GitHub, Jupyter Notebook, MySQL, Excel, MS PowerPoint, MS Word, Solid Edge, Tableau, Power BI

PROJECTS

Full-Stack AI Concierge Agent & API:

Github link: https://github.com/monispamacc-ops/agentic-concierge

- **Overview:** Architected a high-throughput, intent-driven conversational API platform to process unstructured customer queries, extract multi-variable parameters, and drive real-time constraint-based data retrieval.
- **LLM & Natural Language Processing:** Leveraged **Groq's Llama-3.3-70b** via **LangChain** to parse complex, unstructured user prompts into standardized schema attributes (transaction criteria, service types, and numerical budget thresholds) with high reliability.
- **Data Validation & Risk Mitigation:** Engineered strict schema validation and exception-handling pipelines using **Pydantic** to sanitize incoming variables and parse multi-currency regional modifiers (e.g., dynamic scaling and currency formatting e.g., INR/USD formatting) to ensure clean downstream data integrity.
- **Modular Tool-Routing & Dashboarding:** Developed modular mock-data extraction scripts to filter and report multi-constraint datasets, visualized through an interactive **Streamlit** dashboard featuring session tracking and execution auditing logs for operational transparency.
- **Tech Stack:** Python, FastAPI, Pydantic, LangChain, Groq API (Llama-3.3-70b), Streamlit, Git

Customer Intelligence & Predictive Churn Analytics Platform:

Github link: https://github.com/monispamacc-ops/customer-analytics-platform

- **Overview:** Engineered an end-to-end customer portfolio intelligence and behavioral analytics platform to maximize user lifetime value, mitigate attrition risk, and optimize marketing ROI through data-driven predictive modeling.
- **Behavioral Segmentation & RFM Analysis:** Applied unsupervised learning and RFM (Recency, Frequency, Monetary) modeling across 540,000+ transaction records (achieving a Silhouette Score of 0.336) to segment high-value customer cohorts and target personalized engagement strategies.
- **Predictive Churn Modeling & Risk Mitigation:** Developed a supervised **Random Forest** classification pipeline yielding **99% test accuracy**, identifying transactional "Recency" as the primary risk driver (49% feature weight) to enable proactive, data-backed retention campaigns.
- **Product Recommendation Engine:** Built a scalable item-to-item discovery pipeline mapping 4,200+ products to 4,300+ customers utilizing **Cosine Similarity** vector matrices to drive targeted cross-selling and revenue growth.
- **Tech Stack:** Python, Pandas, Scikit-Learn, K-Means Clustering, Random Forest, Cosine Similarity, Git

CineCircle: Cloud-Connected Movie Intelligence & Social Discovery Platform:

Github link: https://github.com/monispamacc-ops/CineCircle

- **Overview:** Architected a high-availability, cloud-connected movie intelligence and social discovery platform designed to optimize relational data management and real-time watchlist orchestration.
- **Database Architecture & Data Integrity:** Designed and normalized a secure relational database schema hosted on **Aiven MySQL Cloud**, implementing strategic indexing to ensure high-performance querying, data consistency, and low-latency response times for concurrent multi-user sessions.

- **Backend Data Pipelines & Query Optimization:** Developed high-performance relational join and data extraction pipelines to synchronize asynchronous frontend user interactions with backend relational data stores, minimizing transaction latency.
- **Cloud Deployment & Version Control:** Orchestrated automated CI/CD deployment pipelines via **GitHub** integration and **Streamlit Cloud**, establishing reliable version control, automated environment provisioning, and zero-downtime application delivery.
- **Tech Stack:** Python, Streamlit, Aiven MySQL Cloud, SQL, Relational Modeling, Git, Streamlit Cloud

WORK EXPERIENCE

In-plant Intern | Hyundai Motor India Limited (June 1, 2026 - June 15, 2026)

- Completed a two-week industrial internship, gaining hands-on exposure to automotive manufacturing processes and plant operations.
- Observed and studied large-scale assembly line workflows and quality control standards in a real-world industrial environment.

Lab Chemical Safety Intern | Department of Chemistry, IIT Hyderabad (Aug '25 - Nov '25)

- Collaborated with faculty and team members to develop a new laboratory manual for the institute.
- Drafted detailed safety procedures and chemical handling guidelines to ensure a secure research environment.
- Documented standard lab practices to standardize safety protocols for students and researchers.

RELEVANT COURSES

Certifications:

- **30 Days Masterclass in Data Analytics** | Pantech.ai (Issued: July 2026)
- **Digital Product Management** | Alison (Issued: June 2026)
- **Diploma in MySQL and Statistics for Data Analysis** | Alison (Issued: May 2026)
- **Basics of UI UX Strategy** | UniAthena (Issued: June 2026)
- **Machine Learning Certification Course for Beginners** | Analytics Vidhya (Issued: July 2026)

Courses:

- **Mathematics:** Probability, Linear Algebra, Differential Equations, Calculus I & II
- **Computer Science:** Data Structures and Applications
- **Mechanics:** Thermodynamics, Fluid Mechanics, Heat and Mass Transfer

POSITIONS OF RESPONSIBILITY

Coordinator, Hospitality | Elan & nvision (May '25 - May '26)

- Worked with a team of 12 members to oversee hospitality operations for the annual fest, ensuring seamless guest management and accommodation services.
- Managed end-to-end logistics and coordinated with vendors to optimize resource allocation, directly improving the experience for over 200 participants from various institutes.

Coordinator, Operations & Logistics | International Relations Cell (Jun '25 - Apr '26)

- Worked with a team of 9 members to execute logistics for international events, guest lectures, and delegation visits.
- Collaborated across IR Cell domains to provide on-ground support and facilitate campus tours for international delegates and visitor.

Coordinator, Ideation & Operations | IIT Hyderabad Alumni Association (Jun '25 - Apr '26)

- Organized alumni meets and helped plan event themes and discussion topics.
- Invited alumni to give talks and hosted on-campus events like podcasts.

EXTRACURRICULAR

Sports: Represented the department in Girls' Cricket and Carrom at inter-departmental tournaments, demonstrating team collaboration and competitive spirit.

NSS Volunteer: Actively contributed to various campaigns at IIT Hyderabad under the Swachh Bharat Mission, promoting environmental sustainability and cleanliness initiatives.

Dance: Performed in various university cultural events and showcases.